Cropping Intensity

Last 5 year previous year question

1. _____ refers to the intensification of cropping in sequence on the same field per year, the succeeding crop is planted after the preceding crop has been harvested, crop intensification is only in the time dimension, there is no intercrop competition. (2022)

a. Intercropping
b. Ratoon cropping
c. Relay cropping
d. Companion cropping

Ans- Relay cropping

CROP INTENSIFICATION

Cropping intensity –

The System of Crop Intensification (SCI) refers to an increase in agricultural production per unit of inputs. **The input includes labour, land, time, fertilizer, seed, feed or cash**. **The aim is to achieve higher output with less use of or less expenditure on land, labour, capital, and water**.

Crop intensification technique includes intercropping, relay cropping, sequential cropping, ratoon cropping, etc.

Objectives

- Maximum **use of land.**
- To decrease **the insect, pest and disease**.
- To increase **the soil fertility and productivity**.
- To **recover the crop loss for** different dormant condition.

- To increase **mixed or inter crop production**.
- To **maximum utilization of solar radiation**.
- To **increase intensification** index by multiple cropping.

Cropping intensity refers to raising of a number of crops from the same field during one agricultural year; it can be expressed through a formula

Cropping Intensity = Gross Cropped Area / Net Sown Area x 100

Or

Area under kharif + rabi + Zaid over area under actual cultivation x 100

$$\textbf{Cropping intensity \%} = \frac{\textbf{Gross cropped area}}{\textbf{Net cultivable area}} \times 100$$

Gross sown area is the cumulative area (sown more than once) which was cultivated within a year by various crops.

For e.g. If a farmer has 100 ha area of land and in the kharif season, he cultivated 90 ha of area and in Rabi 40 ha area and in Zaid 10 ha of area.

Thus, Gross sown area = 150 acre, and net cultivable area is 100 acres

Hence 150/100 =1.5

which means that the farmer utilized the total cultivated land 1.5 times or 150% which is termed as the cropping intensity.

- India has the largest cultivated land in the world, **nearly 51% under cropping, 20% under forest, 6% wastelands and 4-5% pasturelands**. **The potential of increase in the total cultivable area is low**. However, the demand of food crops and industrial crops has been ever increasing. Thus, the only method to increase output is to increase cropping intensity.
- **India had nearly 135% cropping intensity in 2001. But at present it is estimated as 145%**. In comparison to European countries like UK, Italy (~ 190%),

Holland (230%), cropping intensity of India is low. But as compared to Brazil, China and Argentina it is higher.

Methods To Increase Cropping Intensity

Following are the methods/inputs **which are implemented for increasing the cropping** intensity of a region:

- ✓ **Irrigation facilities**.
- ✓ **HYV of crops** which have a short growth period.
- ✓ Cropping methods like **relief farming, mixed farming, strip cropping**
- ✓ **Use of modern inputs like c**hemical fertilizers, pesticides, weedicide
- ✓ **Farm Mechanisation**
- ✓ **Conservation method** – soil/water conservation.
- ✓ **Commercialization,** in trade, capitalistic farming

An increase in the cropping intensity would give more cash/buying power to farmers enabling them to cultivate more.

Cropping Intensity in India has spatial variations and there are 5 derivations for it:

1. Cropping Intensity tends to be higher where labour – intensive farming is found. E.g. West Bengal and Bihar have higher Cropping Intensity because of high labour supplies and unemployed people work on land for livelihood.

2. Subsistence farming has higher crop-diversity and thus, Cropping Intensity of regions with high crop-diversity has greater Cropping Intensity. E.g. Northern Bihar, eastern UP, Malabar west.

3. The climatic control on Cropping Intensity is also palpable, for e.g. in wet areas, Cropping Intensity is higher.

1. Cropping Intensity is **directly proportional to the use of modern inputs, and technology**. It can be shown by comparing the cropping intensity of the following states given in the table below (here Punjab has the highest mechanization which is reflected in its cropping intensity)

Table 1- Cropping Intensity Of Different States In India

State	Cropping Intensity
Punjab	196%
Haryana	188%
West U.P	174-175%
Tamil Nadu	174-175%
West Bengal	170%
Bihar	165%

In the rich alluvial soil regions with higher land capability, Cropping Intensity can be enumerated below:

- River valleys and deltas have high Cropping Intensity due to assurance of water and fertility of soil.
- High population density regions have high Cropping Intensity.
- Regions with high land-man ratio have low Cropping Intensity.
- Government policies related to subsidies; infrastructure facilities also decide Cropping Intensity.

Spatial Pattern of Cropping Intensity

India can be **divided into four regions/zones based on the cropping intensity**. The different zones shown in table are discussed below:

Region I: Very High Cropping Intensity

- This region includes the states of **Punjab, Haryana and West Bengal.**
- Punjab and Haryana plains are **sub-humid alluvial** regions with **good alluvial soil** & moderate to high land capability.
- **Hydrology is sufficient to support the required** cropping pattern of the region.
- After the green revolution, **there was the considerable increase** in the use of modern input and technology

- **Developed rural infrastructure, command area development**, govt incentives and finance, capitalistic farming and larger land holding with farm mechanisation is also the feature of this region.
- **From wheat monoculture, this region has diversified the cropping pattern and along with wheat, rice, cotton, sugarcane, oilseeds, grams are also cultivated.**
- Productivity is higher, agricultural output, per-capital yield and per-capital money income from land is highest.

Table 2- Classification Of Country In 4 Zones On The Basis Of Cropping Intensity

Zone	Cropping Intensity	Cropping Intensity Index
I	Very high Cropping Intensity	>175
II	High Cropping Intensity	150-175
III	Moderate Cropping Intensity	125-150
IV	Low Cropping Intensity	<125

Source- (Statistical survey of India uses this index)

Region II: High Cropping Intensity

- This region includes the **areas of UP, Bihar, Assam, Kerala and Tamil Nadu**
- These regions have **humid climatic conditions with good fertile soil** which makes the land capability high.
- This area comes under land class I which can be cultivated 3 times a year without land fertility getting detoriated.
- Hydrology supports the **agriculture throughout the year with renewal of soil**, almost each year.
- Partial use of **modern inputs and technology, partial mechanisation** has made the agriculture in the region sustainable.
- **The carrying capacity of land is very high;** however, productivity is less as compared to Tamil Nadu and Punjab due to social factors which includes high population density, low land man ratio and subsistence farming which has resulted to high Cropping Intensity.

- Major Crops are rice, wheat, pulses, oilseeds, maize. There is greater diversification due to subsistence nature of agriculture.

Region III: Moderate Cropping Intensity

- This region includes the **areas of Karnataka, Andhra Pradesh, Maharashtra and Madhya Pradesh.**
- These regions **have Sub-humid to semi-arid climatic condition**.
- The soil capability is moderate and after irrigation it can provide greater yield.
- Land man ratio is moderate to high; the carrying capacity of land is low.
- There is need of expansion of dry land agriculture.
- **Major crops grown are Millets, groundnuts, tobacco, and oilseeds.**
- Where irrigation is possible in **river valleys and deltas Cropping Intensity is 190**, e.g. Krishna Godavari delta, tube well irrigated region of Maharashtra, Kaveri basin of Karnataka.
- Productivity of this **region is lower than Class I and Class II states divided on the basis of per hectare yield.**

Region IV: Low Cropping Intensity

- This **region includes the areas of Gujarat, Jammu and Kashmir Rajasthan, North Eastern States, Orissa, Himanchal Pradesh and Jharkhand.**
- These states have physiographic or climatic constraints. **Thus, land capability is classified as 5 to 8 for these states.**
- These **states have highly diversified** cropping pattern.
- Also, the socio-economic **factors like tribal economy and environmental hazards** like **landslides, excessive rainfall and cloud bursts are the key constrains in the development of agriculture of** the region.
- **Productivity is low for these states**. The per-capita agricultural labour output and the total money value earned from agriculture are low.
- The **per-hectare yield of the region is moderate to poor.**